



DEPARTMENT OF ELECTRICAL ENGINEERING AND INFORMATION TECHNOLOGY
POLYTECHNIC SCHOOL OF BASIC SCIENCES

MASTER'S IN AUTOMATION AND ROBOTICS ENGINEERING

ANALYSIS AND CONTROL OF AN ELECTRONIC THROTTLE BODY SYSTEM

Professor:

Prof. Mario di Bernardo

Candidate:

Giuseppe Arena
Matr. P38000311

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- 1 Context and Physical Model
- 2 Open-loop Analysis
- 3 Control Synthesis
- 4 Robustness Analysis
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System Description: The Electronic Throttle Body (ETB) is a valve that regulates air mass flow in order to ensure the stoichiometric ratio in propulsion systems. It is modeled as a rotational mass-spring-damper driven by a DC motor.

Non-linearities:

- **Stiffness:** Cubic hardening spring ($k_1\theta + k_2\theta^3$).
- **Damping:** Non-linear aerodynamic damping ($b_2\dot{\theta}|\dot{\theta}|$).

Mathematical Model:

$$J\ddot{\theta} + b_1\dot{\theta} + b_2\dot{\theta}|\dot{\theta}| + k_1\theta + k_2\theta^3 = u$$

State Space ($x_1 = \theta - \theta_{eq}$, $x_2 = \dot{\theta}$):

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \frac{1}{J}(u - b_1x_2 - b_2x_2|x_2| - k_1x_1 - k_2x_1^3) \end{cases}$$



Figure: Electronic Throttle Valve

Open-Loop Analysis

Equilibrium of the system are found solving for $\dot{x} = 0$.

Since $k_1, k_2 > 0$, the only real equilibrium is the **Origin** (0,0).

Phase Portrait: Evaluating the eigenvalues of Jacobian at origin, the phase portrait highlights an always stable point ($\tau < 0$), while topology depends on parameters due to variations of $D = \tau^2 - 4\Delta$

- Nominal case: **Stable Focus**.
- Increasing of b_1 : **Stable Node**.
- Decreasing k_1 : **Stable Node**.

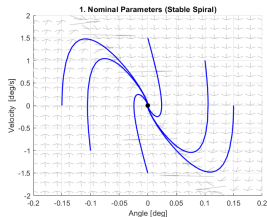


Figure: Nominal Params

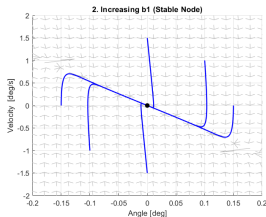


Figure: Increasing B1

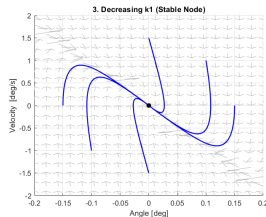


Figure: Decreasing K1

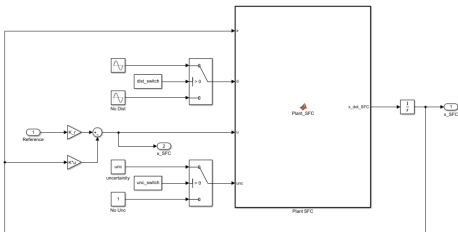
Control Problem Formulation

Objective: Track a desired reference position θ_{ref} minimizing error $e(t)$ fulfilling the following **performance specifications**:

- Settling time: $T_s \leq 0.2s$
- Overshoot: $S\% \leq 13\%$
- Zero Steady-State Error.

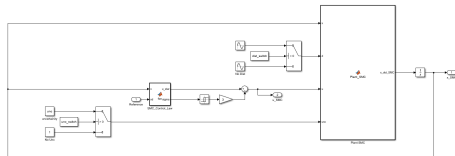
State Feedback (SFC)

Linear strategy on linearized model



Sliding Mode (SMC)

Nonlinear robust strategy



Design:

- Linearization around equilibrium.
- Control law: $u = -Kx + K_r r$.
- K via Pole Placement, K_r for static gain.

Nominal Results:

- Fast rise time ($t_r \approx 0.03s$).
- Overshoot.
- **Issue:** Non-zero steady-state error due to uncompensated non-linearities (cubic spring).

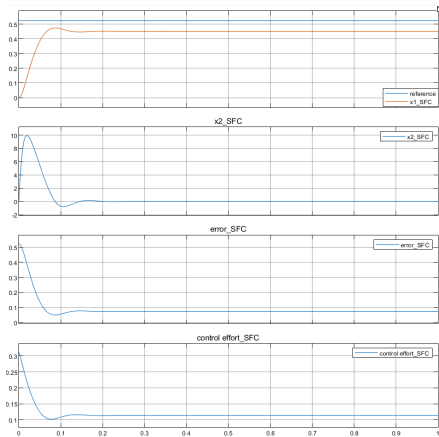


Figure: SFC Nominal Response.

Nonlinear Control: Sliding Mode (SMC)

Design:

- Surface: $\sigma = p_1(x_1 - x_d) + p_2 x_2$.
- Control: $u = u_{eq} + u_{sw}$.
- u_{eq} : Cancels non-linear dynamics.
- $u_{sw} = -K \text{sgn}(\sigma)$: enforces sliding.

Performance	SFC	SMC
Rise Time (T_r) [s]	≈ 0.03	≈ 0.11
Settling Time (T_s) [s]	≈ 0.11	≈ 0.20
Overshoot ($OS\%$)	4%	0%
Steady-State Err (e_{ss})	$\approx 15.0\%$	0%

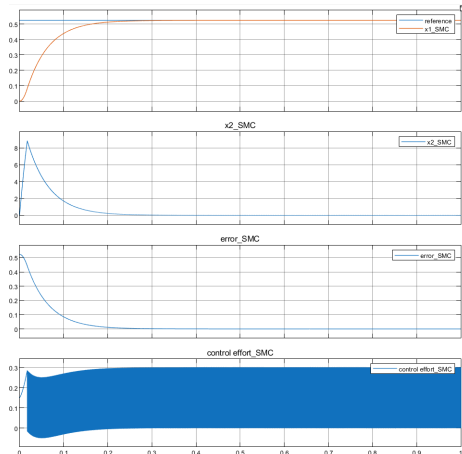
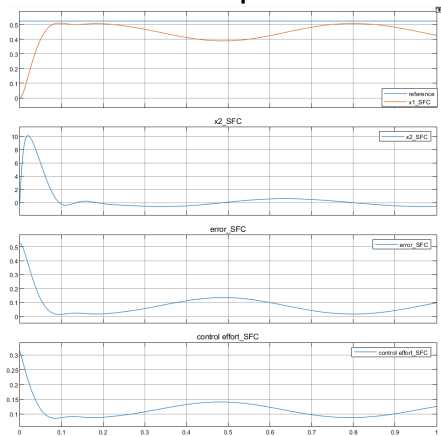


Figure: SMC Nominal Response.

Robustness to Disturbances (Matched)

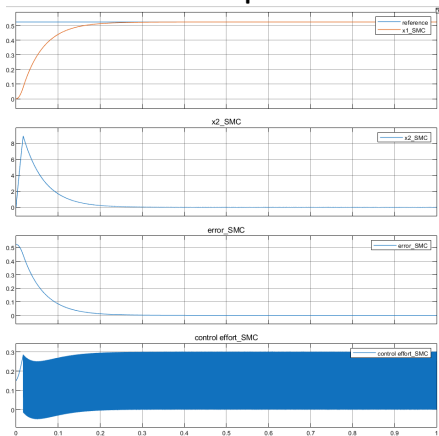
Scenario: Sinusoidal disturbance added to control input ($u + d$).

SFC Response



Failed: Oscillations propagate to output.

SMC Response

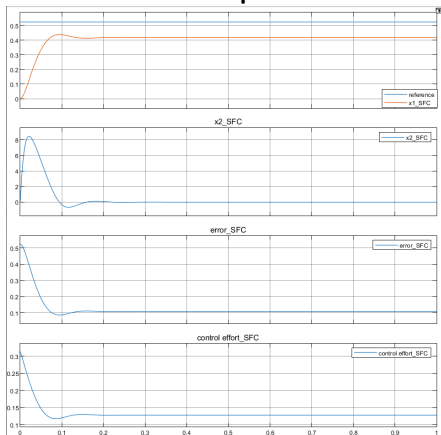


Passed: Complete disturbance rejection.

Robustness to Parametric Uncertainties

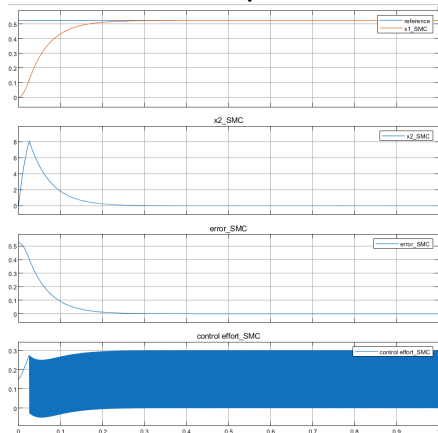
Scenario: 30% parameters uncertainty.

SFC Response



Failed: Large steady-state error due to model mismatch.

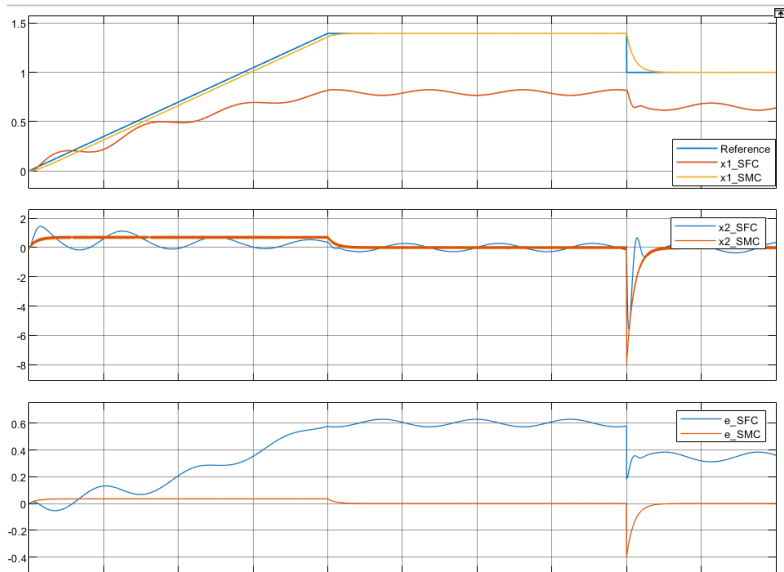
SMC Response



Passed: Robust performance maintained.

Comparison: Worst-Case Scenario

Scenario: Dynamic Reference + Disturbance + Uncertainty.



Analysis Summary

The ETB is a strictly non-linear system. Linear approximations (SFC) are valid only for small deviations and nominal parameters while the Sliding Mode Control (SMC) performs with robustness at every scenario requiring a consistent control effort to actuators.

Trade-off Analysis:

- **SFC:** Simple implementation, smooth control signal (good for actuators), but lacks robustness.
- **SMC:** Superior performance. Guarantees zero error even with heavy uncertainties and disturbances.

Critical Issue & Solution: The main drawback of SMC is **Chattering** (high frequency switching seen in control plots).

- *Risk:* Overheating motor, mechanical wear.
- *Future Work:* Implementation of "Boundary Layer" technique.

Thank You for Your Attention